SMART INDIA HACKATHON 2025

- **Problem Statement ID –** SIH25117
- **Problem Statement Title-** Student Innovation: Swadeshi for Atmanirbhar Bharat - Robotics and Drones
- **Theme-** Robotics and Drones
- **PS Category-** Hardware
- **Team ID-** 69774
- **Team Name-** TransitVision

SAVE-AIR

(Search and Aid Via Emergency Aerial Response)

Proposed Solution :

Core Purpose-

- Identifies Survivors with help of AI and provide basic medical supplies in remote areas.

AI Capabilities -

- 1. Real time navigation and obstacle avoidance
- 2.Return to Home feature
- 3.AI Mapping: Generates live 3D maps of disaster zones to guide safe movement.
- 4.Change Detection: Instantly spots damage by comparing pre- and post-disaster data.

Basic Capabilities -

- Can operate on meshing without external network.
- Identifies the survivors in any disaster using RGB camera ,
- thermal sensors and TWR sensor with the help of highly trained AI
- Can carry basic supplies like Medications .
- Identifies toxic gas leaking or water contamination in disaster zones
- Survey of land and map erosion due to calamity Payload Capacity of 5 Kg approx.

TECHNICAL APPROACH

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Open CV

Computer Vision

Computer vision library for image and video processing.

Gazebo

Simulation Environment

Simulates robots, sensors, and environments realistically in 3D.

Python

Automation-Core

The automation brain—converting your concepts into intelligent, scalable execution.

RTAB Map

SLAM

Simultaneously localizing a sensor while mapping its surrounding environment.

TensorFlow

Meachine Learning

Used for AI-based victim detection from camera or thermal sensor images.

ROC

Robort Oprationg S/M

Communication between sensors, cameras, and control algorithms for smooth drone operation.

FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

- Carbon-fiber composite frame ~25-40% weight reduction high durability in rugged/disaster zones.
- Dual RGB + Thermal Effective in low visibility, smoke, nighttime, plus opt extra sensors (LIDAR)
- Build autonomy on open-source platforms (ArduPilot /PX4) for nort-orrays

CHALLENGES

- Battery endurance 4 Swappable packs + solar docking / recharge
- Data Zeroization: Securely and permanently overwriting or crypto-erasing sensitive information.
- Communication in remote zones. Mash networking/relay drones, satellife uplinks

VIABILITY & IMPACT

- Enables rapid victim detection & damage assessment in remote /disaster areas
- Platform re-uusable disaster relier, forest fire monitoring, environmental mapping
- Sustainable cost modular design, local manufacturing

IMPACT AND BENEFITS

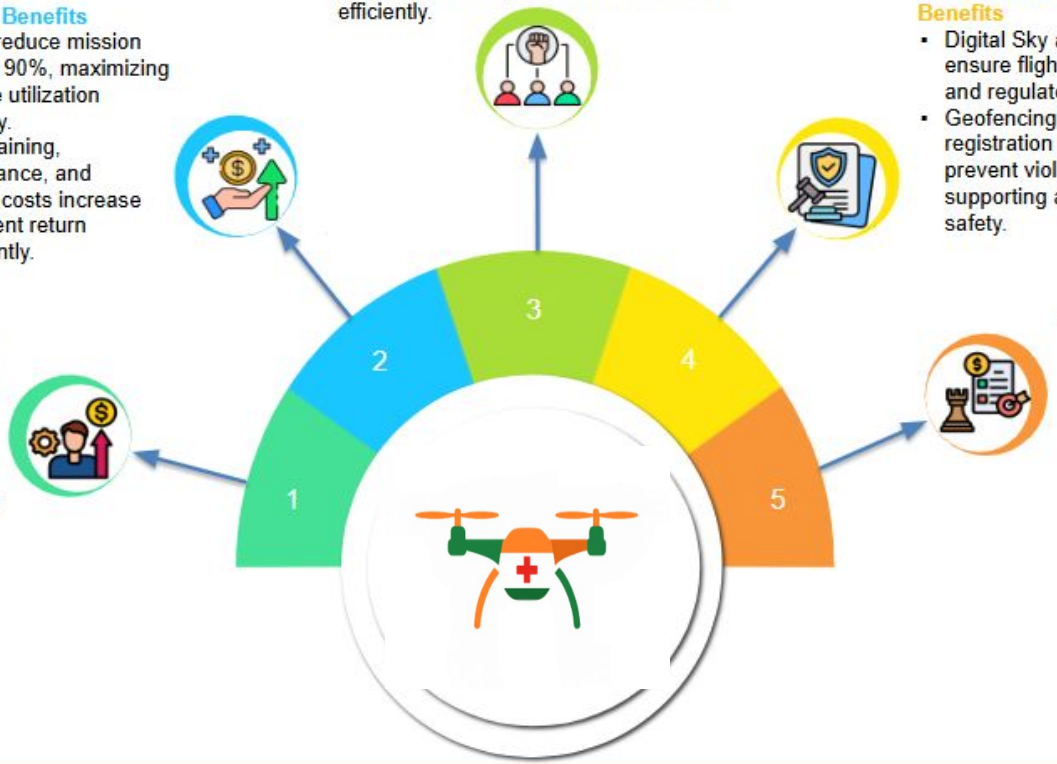
Social Impact
- Bridges rural emergency care gap; 65% Indians face access issues.
- Medicine from Sky delivered 10,000 products, reaching isolated communities efficiently.

Economic Benefits
- Drones reduce mission costs by 90%, maximizing resource utilization efficiency.
- Lower training, maintenance, and logistics costs increase investment return significantly.

Compliance & Safety Benefits
- Digital Sky and NPNT ensure flights are safe and regulated.
- Geofencing, auto registration systems prevent violations, supporting airspace safety.

Operational Benefits
- AI drones offer instant aerial surveillance and disaster zone mapping accuracy.
- Real-time sensors enable 30% faster decisions for life-saving response.

Strategic Benefits
- Backed by incentives, India aims for global drone technology leadership.
- Supports disaster planning, local manufacturing, quick recovery, disaster readiness nationwide.



RESEARCH AND REFERENCES

Research

- Research89% of emergency response teams reported significant time reduction using AI-assisted drones for search and rescue operations in disaster-affected areas.
- Over 75% of medical professionals identified drone-delivered emergency supplies as critical for reducing response times in rural and remote locations.
- Initial deployment challenges were noted by 31% of stakeholders regarding weather dependency and flight range limitations for autonomous systems.
- 92% of thermal imaging accuracy in life detection during search operations would dramatically improve survivor location capabilities in disaster zones.
- Phase 1 rescue teams showed 78% support for mesh-networked drone swarms to coordinate multi-area emergency response and real-time communication during disasters.

References

- Use of UAVs in disaster relief-Nepal Earthquake (2015) –https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4833776/
- Kerala Floods (2018) – https://www.wired.com/story/kerala-floods-drones/
- WHO Guidelines on emergency supply delivery – https://www.who.int/publications/i/item/who-emergency-medical-supplies
- IEEE Paper on AI UAV navigation – https://ieeexplore.ieee.org/document/8723204
- NDRF Disaster Response – https://ndrf.gov.in/
- UNDRR Disaster Framework – https://www.undrr.org/publications
- ISRO Remote Sensing Applications – https://www.isro.gov.in/EOS.html